1. $M = (\{q_0, q_1, q_2, q_3, q_4\}, \{a, b\}, \{z, A\}, \delta, q_0, z, \{q_4\})$

$\delta(q_0, a, z) = \{(q_1, z)\}$
$\delta(q_0, a, A) = \{(q_1, A)\}$ } 第1個 a

$\delta(q_1, a, z) = \{(q_2, z)\}$
$\delta(q_1, a, A) = \{(q_2, A)\}$ } 第2個 a

$\delta(q_2, a, z) = \{(q_0, AAz)\}$
$\delta(q_2, a, A) = \{(q_0, AAA)\}$ } 第3個 a

$\delta(q_0, b, A) = \{(q_3, \lambda)\}$ 第1個 b

$\delta(q_3, b, A) = \{(q_3, \lambda)\}$ 第2個 b

$\delta(q_0, \lambda, z) = \{(q_4, \lambda)\}$ → $n=0$ 時
$\delta(q_3, \lambda, z) = \{(q_4, \lambda)\}$ → 比對完成

2. $S \to z_{00} \mid z_{01} \mid z_{02}$

$z_{0k} \to a A_{0m} z_{mk}$ for all $m, k \in \{0, 1, 2\}$

$A_{0k} \to a A_{0m} A_{mk}$ for all $m, k \in \{0, 1, 2\}$

$A_{01} \to b$ (pop)

$A_{11} \to b$ (pop)

$z_{12} \to \lambda$ (pop)

Equivalent CFG:

$S \to z_{02}$

$z_{02} \to a A_{01} z_{12}$

$A_{01} \to a A_{01} A_{11} \mid b$

$A_{11} \to b$

$z_{12} \to \lambda$ #

3. $M = (\{q_0, q_1, q_2\}, \{a, b\}, \{S, A, B, a, b, z\}, \delta, q_0, z, \{q_2\})$

$\delta(q_0, \lambda, z) = \{(q_1, Sz)\}$

$$\left.\begin{aligned} &\delta(q_1, \lambda, S) = \{(q_1, aAb), (q_1, bBa)\} \\ &\delta(q_1, \lambda, A) = \{(q_1, B), (q_1, a)\} \\ &\delta(q_1, \lambda, B) = \{(q_1, bB), (q_1, b)\} \end{aligned}\right\}$$ 展開（遇到變數）

$$\left.\begin{aligned} &\delta(q_1, a, a) = \{(q_1, \lambda)\} \\ &\delta(q_1, b, b) = \{(q_1, \lambda)\} \end{aligned}\right\}$$ 比對（遇到終結符號）

$\delta(q_1, \lambda, z) = \{(q_2, \lambda)\}$